Local–Global Sparse Transformer for Road Extraction From Remote Sensing Imagery

Xu Wang, Luyao Wang, Wenxin Huang, *Member, IEEE*, Yibo Liu, Sheng Zhang, and Xian Zhong, *Senior Member, IEEE*

***Abstract*—Accurate road extraction from high-resolution remote sensing imagery is crucial for applications such as road network generation, urban planning, autonomous driving, and military operations. Although deep learning has significantly advanced segmentation performance, road extraction remains challenging because roads exhibit geometric and structural characteristics that differ markedly from general objects. We propose a local–global sparse transformer network (LGST-Net) that leverages sparse attention (SA) to capture both local details and global context. First, a coarse-to-fine feature-enhanced preprocessing (C2F-FEP) module extracts low-level features at a fine-grained scale during the model's initial stage. Next, we design the LGST backbone, which comprises three novel components: local shift-mask SA (LSM-SA), global compressed sparse attention (GC-SA), and a spatial-channel gate. Extensive experiments on DEEPGLOBE and ROADTRACER datasets demonstrate that LGST-Net outperforms state-of-the-art methods. Our code will be publicly available at https://github.com/JaymeWX/Road_LGST-Net**

***Index Terms*—High-resolution imagery, remote sensing imagery, road extraction, shift-window attention, transformer-based deep learning.**

I. INTRODUCTION

ROADS are indispensable elements in remote sensing imagery, particularly in urban environments. Accurate road detection underpins applications such as urban planning, traffic optimization, and autonomous navigation [1], [2]. However, significant variations in road texture, illumination, and weather conditions complicate this task.

Recent studies [3], [4] treat road detection as a pixel-wise classification problem, segmenting an image into road and background. Methods broadly fall into two categories: manual feature-based techniques [5], [6], which rely on hand-crafted features and mathematical models (e.g., *K*-means clustering, conditional random fields (CRFs) [5]), and deep learning-based approaches [7], [8]. Although traditional methods are computationally efficient, they often lack generalization across diverse datasets.

The advent of convolutional neural networks (CNNs) has driven remarkable progress in vision tasks [9], [10], [11], [12], including road extraction [4]. Encoder–decoder frameworks, especially U-Net variants [7], [13], [14], [15], [16], leverage CNN or ResNet backbones to refine feature representations. More recently, self-attention mechanisms and vision transformers (ViTs) have succeeded in both NLP and computer vision [17], [18], [19], [20]. Integrating ViT modules into encoder–decoder architectures introduces global context, further improving road extraction [21], [22], [23]. However, in very high-resolution (VHR) imagery, global self-attention incurs substantial computational costs.

To address this, many research efforts have focused on sparse attention (SA), a constrained self-attention mechanism that computes pairwise affinities only between selected

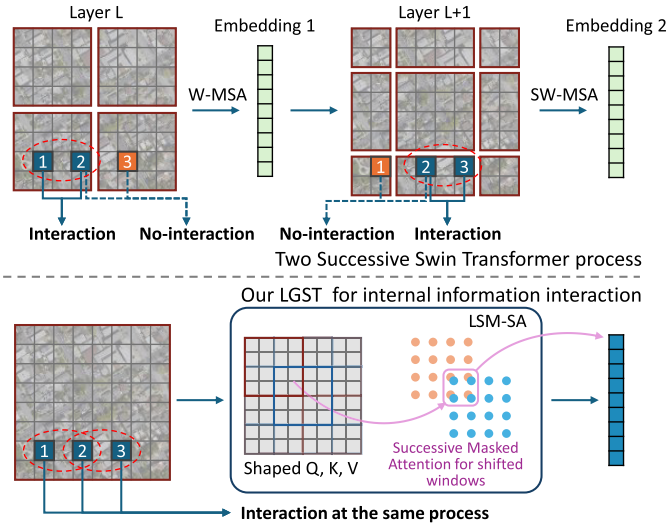


Fig. 1. Motivation for LGST-Net. (Top row) Two successive swin transformer blocks, W-MSA and SW-MSA, reduce computational cost but only allow indirect interactions between neighboring embeddings. In contrast, our one-step LSM-SA applies multiple shifts to a single QKV group, enabling direct cross-window attention and enhanced feature extraction at window boundaries.

Received 26 May 2024; revised 21 September 2024, 20 January 2025, 13 May 2025, and 24 June 2025; accepted 10 August 2025. Date of publication 14 August 2025; date of current version 18 September 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62102423, Grant 42301493, Grant 62301213, and Grant 62271361; and in part by Hubei Provincial Key Research and Development Program under Grant 2024BAB039. *(Corresponding author: Luyao Wang.)*

Xu Wang, Luyao Wang, Yibo Liu, and Sheng Zhang are with the School of Information and Communication, National University of Defense Technology, Wuhan 410073, China (e-mail: wangxu9191@gmail.com; wangluyao@nudt.edu.cn; liuyiboa@163.com; zhangsheng_nudt@qq.com).

Wenxin Huang is with Hubei Key Laboratory of Big Data Intelligent Analysis and Application, School of Computer Science and Information, Hubei University, Wuhan 430062, China (e-mail: wenxinhuang_wh@163.com).

Xian Zhong is with Hubei Key Laboratory of Transportation Internet of Things and the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan 430070, China, and also with the State Key Laboratory of Maritime Technology and Safety, Wuhan University of Technology, Wuhan 430063, China (e-mail: zhongx@whut.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3598813

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:30:11 UTC from IEEE Xplore. Restrictions apply.

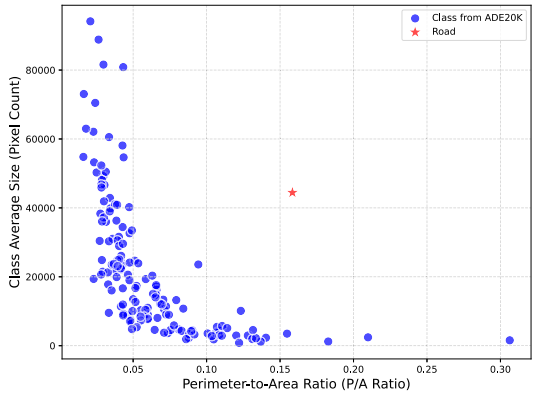


Fig. 2. Pixel-shape distribution comparison. Scatter plot showing the relation between average pixel count and perimeter-to-area ratio across object categories. Blue dots denote 150 ADE20K classes, while red pentagrams indicate road segments in remote sensing imagery.

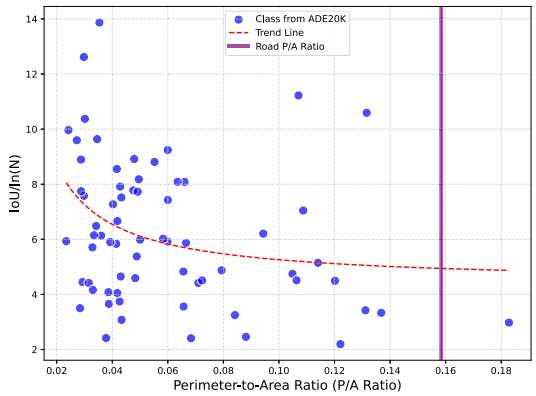


Fig. 3. Segmentation accuracy versus shape complexity. Scatter plot of normalized IoU (IoU/ $\ln N$, where N is the number of samples to mitigate class imbalance) versus perimeter-to-area ratio. Blue dots represent ADE20K categories with IoU > 10%, the dashed line indicates the overall trend, and the purple vertical line marks the ratio for remote sensing road targets.

subsets or aggregated tokens of input embeddings instead of exhaustively evaluating all possible pairs. Liu et al. [24] proposed the swin transformer, which partitions embeddings into nonoverlapping windows and shifts them to enable cross-window information flow. As shown in Fig. 1, successive window multihead self-attention (W-MSA) and shifted W-MSA (SW-MSA) layers allow each embedding to attend to its neighbors, yet the two-step process still yields only indirect interactions.

Remote sensing imagery follows consistent capture patterns, stable satellite orbits, and fixed imaging distances, revealing road networks as large-scale, lattice-like structures with unique geometric signatures. To highlight this structural disparity, we compare natural image objects and remote sensing roads in Figs. 2 and 3. In natural images, objects of increasing area tend to exhibit decreasing perimeter-to-area ratios, and vice versa. As shown in Fig. 2, most natural categories cluster along this inverse trend, reflecting similar shapes across diverse classes. In contrast, road segments combine large pixel areas with unusually high perimeter-to-area ratios, underscoring their distinct morphology and posing unique segmentation challenges.

The perimeter-to-area ratio effectively distinguishes shape and structural differences among objects of equal area. Detection accuracy in natural scenes tends to decrease as this ratio increases, whereas road segments exhibit markedly higher ratios, highlighting their unique morphology. This analysis reveals two core challenges for road segmentation: 1) sparse, large-scale topology and 2) fine-grained, linear geometry. Since both global connectivity and local precision depend critically on edge accuracy, we term tasks with extremely high perimeter-to-area ratios as pixel-sensitive segmentation tasks (PSSTs).

In this article, we introduce the local–global sparse transformer network (LGST-Net) for road extraction from remote sensing images. As shown at the bottom of Fig. 1, our local shift-mask SA (LSM-SA) applies multiple window partitions and shifts to a single set of query, key, and value embeddings simultaneously, enabling direct, range-enhanced feature interactions at window boundaries. Built on an efficient encoder–decoder framework, LGST-Net is both simple and powerful, and extensive experiments demonstrate its superior performance.

The main contributions of this work are threefold.

1) We introduce the concept of PSSTs to characterize the unique challenges of extracting slender, lattice-like structures, such as roads, in high-resolution remote sensing imagery. To tackle PSST, we develop a coarse-to-fine feature enhancement module that progressively amplifies pixel-level details, ensuring precise representation of narrow, elongated features from the earliest layers of the network.
2) To overcome the prohibitive cost of global attention while preserving its contextual power, we design the LGST. By combining targeted local interactions with a compressed global aggregation mechanism, augmented by an adaptive spatial-channel gating strategy, LGST delivers both fine-scale sensitivity and broad-range coherence in a single, efficient attention block.
3) We integrate these innovations into a unified encoder–decoder framework, LGST-Net, which leverages SA to seamlessly fuse detailed local cues with holistic scene context. This architecture not only streamlines computation but also sets a new benchmark for robustness and accuracy in road extraction from diverse remote sensing datasets.

II. Related Works

A. Road Extraction

In our view, the development of deep learning has driven road extraction through three distinct phases, marked by the emergence of key architectures: CNN, encoder–decoder, and ViT. Representative works from these periods are discussed below.

1) CNN and FCN Approaches: Shelhamer et al. [25] introduced fully convolutional networks (FCN) for semantic segmentation of remote sensing imagery. Zhong et al. [26] applied a CNN to segment roads and buildings, analyzing hyperparameters (learning rate, input size, epochs, and stride)

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:30:11 UTC from IEEE Xplore. Restrictions apply.

for high-resolution images. Wei et al. [27] refined CNNs to encode spatial and geometric road structures, and Alshehhi and Marpu [28] proposed a patch-based CNN for road and building extraction from VHR data.

2) Encoder–Decoder Models: Encoder–decoder architectures effectively project inputs into latent spaces for segmentation. Panboonyuen et al. [29] developed a deep convolutional encoder–decoder (DCED) network using exponential linear units (ELUs) instead of rectified linear units (ReLUs) to improve convergence. SegNet [30] treated road extraction as an image segmentation problem, and Panboonyuen et al. [31] enhanced it with landscape metrics and CRFs. Zhang et al. [7] and Li et al. [32] integrated ResNet backbones into U-Net, while Lu et al. [13] proposed GAMS-Net, employing multiscale residual learning to expand receptive fields. Variants such as D-LinkNet [14], NL-LinkNet [15], and UNet3+ [33] further improved performance. Zhu et al. [34] introduced a global context-aware block to capture long-range dependencies within the encoder–decoder framework.

3) ViT-Based Methods: Since its introduction [35], ViT has excelled in image processing and segmentation. Fu et al. [36] presented the dual attention network (DANet) to combine local features with global context. SegFormer [37] simplified transformer decoders using lightweight MLPs, and Segmenter [38] leveraged pretrained ViT backbones with point-wise linear or mask transformers for decoding. The pyramid ViT (PVT) [39] introduced a hierarchical feature pyramid for downstream tasks. Particularly, in the field of road extraction, attention and ViT have also inspired many valuable research works. Zheng et al. [40] proposed SETR, replacing CNN backbones with ViT for segmentation. Extensions of U-Net, such as CapsUNet and cascaded-UNet, integrate transformer modules into encoder–decoder pipelines for remote sensing [17], [18], [19], [20], [41]. Compared to CNNs, ViTs model global-range pixel relations and feature representation, but global self-attention demands substantial memory and computation, particularly for VHR imagery.

B. SA-Based Methods

To improve efficiency at high resolutions, various ViT variants have been proposed Yuan et al. [42] and Chu et al. [43]. Chen et al. [44], [45] selected a number of the most important spatial and channel tokens to model the long-range dependency of the input image. In addition to filtering, some methods [46], [47] employ token aggregation techniques to compress the number of tokens spatially, thus achieving computational reduction. Among such methods, swin transformer [24] stands out as one of the most impactful approaches. It partitions embeddings into nonoverlapping windows and shifts them to facilitate cross-window interactions, reducing complexity from $O(n^2)$ to $O(n)$. Despite this efficiency gain, interactions remain indirect since attention is computed separately in each shift.

Several works integrate swin modules into encoder–decoder architectures for segmentation. Lin et al. [48] introduced DS-TransUNet by combining swin with U-Net for medical image segmentation. He et al. [49] embedded swin in a dual-encoder U-Net for remote sensing. For road extraction specifically, Liu et al. [50] designed RoadFormer with a swin backbone, and Chen et al. [51] proposed CoSwin, a dual-branch encoder merging swin and ResNet features. Although these methods benefit from reduced complexity, as shown in Fig. 1, they still rely on indirect interactions across shifted windows.

III. Proposed Method

A. Network Architecture

We present the LGST-Net framework; the overall architecture is shown in Fig. 4. The model adopts a standard encoder–decoder design, which is widely used in segmentation tasks for its simplicity and efficiency.

First, low-level features are extracted from the input image via a simple convolutional network. To enhance these features for road-specific structures, we introduce a coarse-to-fine feature-enhanced preprocessing (C2F-FEP) module during the preprocessing stage (details in Section III-B).

The enhanced features are partitioned into nonoverlapping 4×4 patches and flattened into embeddings. The encoder comprises a sequence of LGST blocks interleaved with downsampling modules; each downsampling halves both spatial dimensions, as illustrated in Fig. 4(a). The decoder mirrors this structure with upsampling modules and LGST blocks: at each resolution, the decoder LGST block receives a residual addition of the upsampled features and the corresponding encoder output. Each upsampling doubles the spatial dimensions. Finally, the head module projects the last feature map into a binary road mask. We train the network using a composite loss combining dice loss and binary cross-entropy (BCE); no new loss functions are introduced.

B. Coarse-to-Fine Feature-Enhanced Preprocessing

The C2F-FEP module refines low-level features through a linear combination of channel, spatial, and pixel attention. Similar to CBAM [52], it sequentially applies channel and spatial attention, then adds pixel-level attention to capture the fine-grained details of narrow road structures.

C. Local–Global Sparse Transformer

Fig. 5 illustrates the LGST, which follows the standard MSA of ViT but replaces global attention with our local–global sparse attention (LGSA).

Fig. 6 details the LGSA workflow. We restrict attention to sliding windows that shift by varying strides and incorporate a compressed attention mechanism for global context. Unlike the swin transformer, our approach eliminates embedding reshaping, tensor rolling, and window partition/reverse operations, improving runtime efficiency. The LGSA module comprises three components: LSM-SA, global compressed sparse attention (GC-SA), and a spatial-channel gate.

1) LSM Sparse Attention: We implement shift-window attention via masking. For each QKV group ($Q \in \mathbb{R}^{M\times d}$, $K, V \in \mathbb{R}^{N\times d}$), the attention output at shift step i (stride t_i, window size w) is

$$O_i = \text{softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d}} + \text{Mask}_{t_i,w}\right) O_{i-1} \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:30:11 UTC from IEEE Xplore. Restrictions apply.

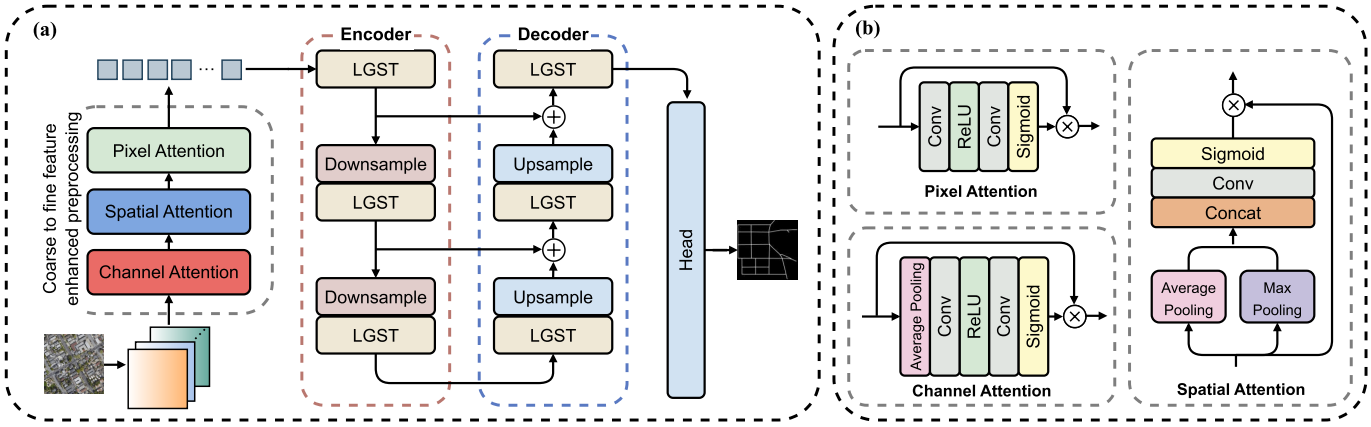


Fig. 4. LGST-Net framework. (a) Overview of the network architecture. (b) Coarse-to-fine feature-enhancement preprocessing, comprising sequential channel attention, spatial attention, and pixel-level attention. The core LGST module is highlighted in (a) and detailed in Section III-C.

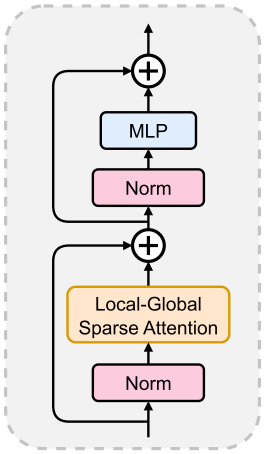


Fig. 5. LGST. Overview of the transformer architecture incorporating the proposed LGSA mechanism.

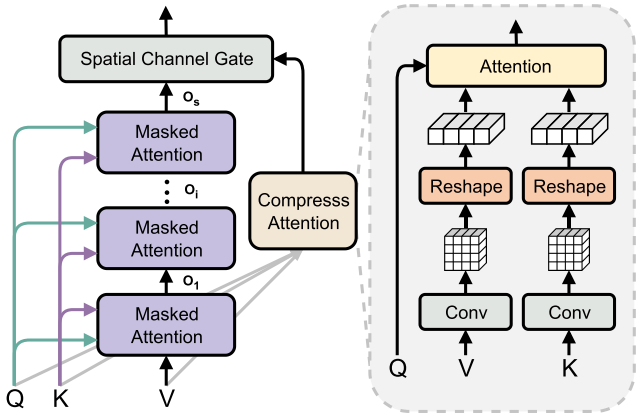


Fig. 6. LGSA workflow. Detailed workflow of the LGSA module.

where $O_0 = V$, and $\text{Mask}_{t_i,w} \in \mathbb{R}^{M\times N}$ is defined as follows:

$$\text{Mask}_{t_i,w}(m,n) = \begin{cases} 0, & \lfloor R(m)/w \rfloor = \lfloor R(n)/w \rfloor \\ & \wedge\ \lfloor C(m)/w \rfloor = \lfloor C(n)/w \rfloor \\ -\infty, & \text{otherwise.} \end{cases} \tag{2}$$

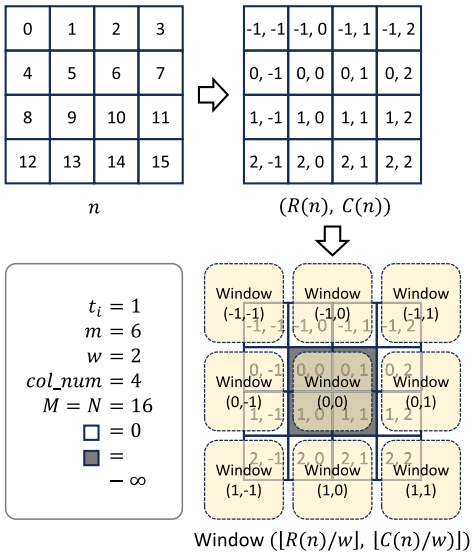


Fig. 7. Mask row construction example. Illustration of the process for constructing a single row of the masking matrix $\text{Mask}_{t_i,w}$.

Here, $m = 0,\ldots,M-1$ and $n = 0,\ldots,N-1$ index the rows and columns of $\text{Mask}_{t_i,w}$. Here,

$$R(x) = \lfloor x/\text{col_num} \rfloor - t_i \tag{3}$$

$$C(x) = (x \bmod \text{col_num}) - t_i \tag{4}$$

convert a linear index x to its row and column coordinates. Masked entries $(-\infty)$ are excluded by the softmax. An example of constructing one row of $\text{Mask}_{t_i,w}$ is shown in Fig. 7.

In practice, we use Triton to implement our efficient sparse computation. We precompute a Boolean mask (true for allowed pairs, false for masked) based on (2). Since this mask depends only on t_i and w, it can be cached and reused, and its high sparsity further accelerates attention via optimized sparse operations. In terms of computational analysis, our LSM-SA module reduces the complexity of standard attention from $O(N^2)$ to $O(N)$.

2) GC Sparse Attention: To capture global context (see Fig. 6), we compress K and V using nonoverlapping

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:30:11 UTC from IEEE Xplore. Restrictions apply.

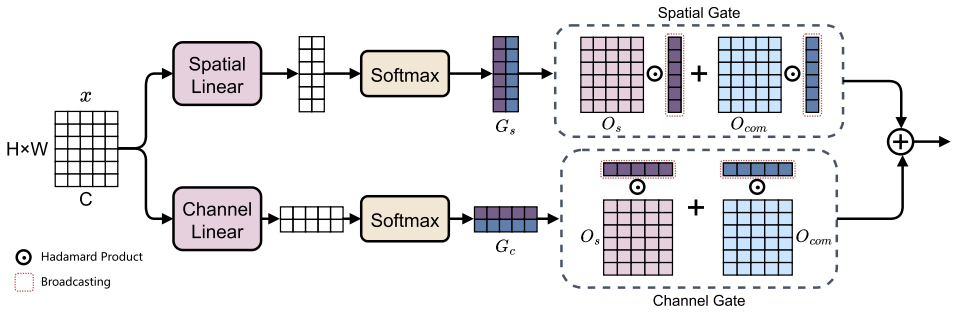


Fig. 8. Spatial-Channel gate workflow. Detailed workflow of the proposed spatial-channel gating module.

convolutional kernels (kernel size = stride), yielding K_{com} and V_{com}

$$Q_{\text{com}} = \text{Conv}\left(\text{Reshape}\left(Q\right)\right) \tag{5}$$

$$K_{\text{com}} = \text{Conv}\left(\text{Reshape}\left(K\right)\right). \tag{6}$$

We then compute

$$O_{\text{com}} = \text{softmax}\left(\frac{QK_{\text{com}}^{\text{T}}}{\sqrt{d}}\right) V_{\text{com}} \tag{7}$$

where O_{com} has the same dimension as O_i in (1). Taking advantage of compressed V and K, our GC-SA module reduces the complexity of standard attention from $O(N^2)$ to $O(kN)$, where $k \ll N$.

3) Spatial-Channel Gate: We propose a spatial-channel gating mechanism to fuse the multistep masked attention and compressed attention outputs dynamically. The detailed flow is illustrated in Fig. 8. Specifically, given input x, we apply two linear projections followed by softmax to obtain spatial weights G_s and channel weights G_c

$$G_s = \text{softmax}\left(\text{SpatialLinear}\left(x\right)\right) \tag{8}$$

$$G_c = \text{softmax}\left(\text{ChannelLinear}\left(x\right)\right) \tag{9}$$

where $G_s \in \mathbb{R}^{M\times 2}$ and $G_c \in \mathbb{R}^{2\times d}$. Each column of G_s provides spatial gating coefficients, and each row of G_c provides channel gating coefficients. The final fused output is

$$O = O_s \odot \text{BC}\left(g_s^1\right) + O_{\text{com}} \odot \text{BC}\left(g_s^2\right) \tag{10}$$

$$+ O_s \odot \text{BC}\left(g_c^1\right) + O_{\text{com}} \odot \text{BC}\left(g_c^2\right) \tag{11}$$

where BC$(\cdot)$ broadcasts the gating vectors to match the dimensions of O_s and O_{com} for element-wise multiplication.

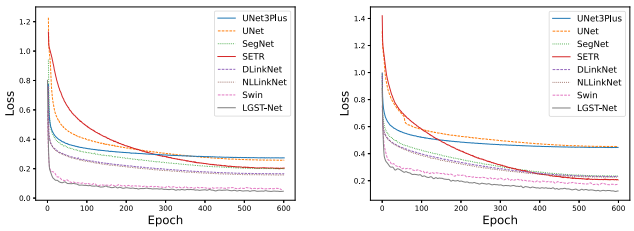
IV. Experimental Analysis

A. Datasets

We evaluate our method on two benchmarks: DeepGlobe [53] and RoadTracer [54].

DeepGlobe comprises 8570 images of size 1024 × 1024 at 0.5 m per pixel, captured over rural and urban areas in Thailand, Indonesia, and India. It is split into 6626 training, 1243 validation, and 1101 test images.

RoadTracer contains 3000 urban-core images (4096 × 4096, 0.6 m resolution), which we tile into four 1024 × 1024 patches, yielding 12 000 samples: 8400 for training, 1800 for validation, and 1800 for testing.



(a) (b)

Fig. 9. Training loss curves. Training loss on DeepGlobe and RoadTracer datasets over successive epochs. (a) DeepGlobe and (b) RoadTracer.

B. Implementation Details

We implement all experiments in PyTorch on an NVIDIA RTX A6000 (48 GB). To fit memory constraints, we resize input patches to 512 × 512 during training. We apply five augmentations per batch (50% chance each): HSV jitter, ±90° rotation, horizontal/vertical flips, translation, and scaling. Training runs for 600 epochs with batch size 16 using AdamW (initial learning rate 1×10^{-5}, weight decay 0.02).

C. Evaluation Metrics

Road extraction is treated as a pixel-wise binary classification task, where road pixels are considered the positive class and background pixels the negative class. We report precision, recall, intersection over union (IoU), and $F1$-score.

D. Training Comparison

Fig. 9 compares training losses on DeepGlobe and RoadTracer. LGST-Net achieves the lowest training loss on both datasets. Its superior test performance (see Section IV-F) indicates minimal overfitting and strong generalization. The higher loss on RoadTracer reflects its more complex urban scenes and annotation variability.

E. Computational Comparison

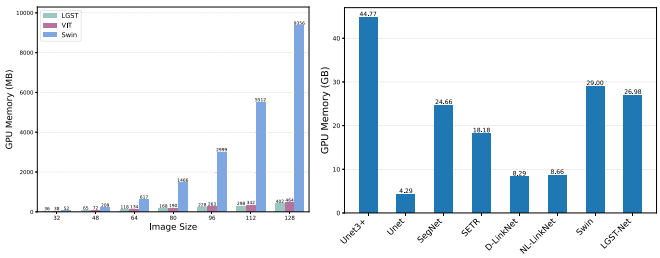
1) Parameter and FLOPs: Table I summarizes parameter counts and FLOPs, evaluated using a batch size of eight and an input resolution of 512 × 512 pixels. UNet3+ [33] has the fewest parameters but the highest FLOPs; SETR [40] has the most parameters. LGST-Net lies in the middle, matching RADANet [58] in parameter count and falling between SGCN-Net [56] and SegNet [30] in FLOPs.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:30:11 UTC from IEEE Xplore. Restrictions apply.

TABLE I

COMPARISON OF MODEL COMPLEXITY AND PERFORMANCE ON DEEPGLOBE AND ROADTRACER. PARAMETERS, FLOPS, AND FPS ARE SHOWN ALONGSIDE RECALL, PRECISION, $F1$-SCORE, AND IOU FOR EACH DATASET. BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY. † INDICATES REPRODUCED METHODS

Model	#Params (M)	FLOPs (G)	FPS	DEEPGLOBE				ROADTRACER			
				Recall	Precision	F1	IoU	Recall	Precision	F1	IoU
UNet† [55]	39.50	**5.92**	**49**	74.00	76.52	75.24	61.93	71.14	71.73	71.44	56.69
SegNet† [30]	29.44	160.64	18	**79.65**	73.15	76.26	63.32	72.50	70.53	71.50	58.44
D-LinkNet [14]	31.10	26.72	37	-	-	-	64.12	69.56	70.24	69.89	56.19
SETR† [40]	229.82	80.66	18	65.06	74.91	69.64	55.13	62.24	67.40	64.72	48.66
SGCN-Net† [56]	42.73	125.00	-	-	-	73.77	57.49	-	-	-	-
MDANet† [57]	31.90	88.60	-	-	-	75.19	60.21	-	-	-	-
UNet3+† [8]	**6.76**	201.66	10	76.69	78.17	77.42	64.48	70.44	69.65	70.04	55.82
NL-LinkNet [15]	21.80	25.18	37	-	-	-	65.00	70.16	70.59	70.37	56.57
RADANet [58]	73.58	212.00	-	-	-	78.72	64.22	-	-	-	-
Swin† [24]	58.94	241.66	9	74.42	79.09	76.69	62.19	72.41	**75.98**	74.15	58.92
LGST-Net	78.36	127.64	22	77.53	**80.70**	**79.08**	**65.40**	**74.09**	74.97	**74.53**	**59.40**



(a) (b)

Fig. 10. GPU memory usage comparison. (a) GPU memory consumption for LGST, ViT, and two successive swin transformer blocks across varying input sizes. (b) Memory usage of all methods using a batch size of 16 and an input resolution of 512 × 512.

2) Training Speed: We report training throughput (FPS) in Table I. Despite its moderate size, LGST-Net matches the speed of SegNet [30] and SETR [40].

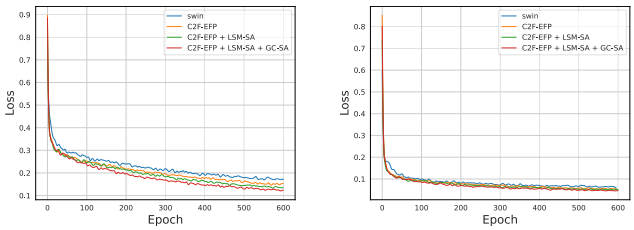
3) GPU Memory Usage: Fig. 10 shows GPU memory usage for LGST-Net, swin transformer, and ViT modules across input sizes. Replacing LGST with global ViT attention causes a memory surge, whereas LGST scales linearly. Overall, LGST-Net maintains a moderate memory footprint.

F. Quantitative Comparison

We compare LGST-Net with UNet3+ [33], UNet, SegNet [30], SETR [40], D-LinkNet [14], and NL-LinkNet [15]. Table I reports precision, recall, $F1$-score, and IoU on both datasets. On DEEPGLOBE, LGST-Net achieves the highest precision (80.7%), $F1$-score (79.08%), and IoU (65.40%), a 0.4 pp IoU gain over NL-LinkNet. On ROADTRACER, it leads with recall (74.09%), $F1$-score (74.53%), and IoU (59.40%). Performance is consistently better on DEEPGLOBE, reflecting the greater complexity of urban-core road networks. These results confirm that dense urban scenes pose greater challenges than rural and suburban areas, and demonstrate LGST-Net's robust, state-of-the-art performance.

G. Ablation Experiments

We perform ablation studies to assess the impact of each LGST-Net component. Four configurations are compared:



(a) (b)

Fig. 11. Ablation loss curves. Training loss progression for ablation studies on (a) ROADTRACER and (b) DEEPGLOBE datasets over successive epochs.

TABLE II

ABLATION STUDY ON ROADTRACER. IMPACT OF THE C2F-FEP, LSM-SA, AND GC-SA MODULES ON ROAD EXTRACTION PERFORMANCE. BEST RESULTS ARE HIGHLIGHTED IN BOLD

Modules			Road IoU	F1-score	Precision	Recall
C2F-FEP	LSM-SA	GC-SA	(%)	(%)	(%)	(%)
○	○	○	58.92	74.15	75.98	72.41
●	○	○	59.18	74.36	**76.47**	72.35
●	●	○	59.29	74.44	75.48	73.44
●	●	●	**59.40**	**74.53**	74.97	**74.09**

TABLE III

ABLATION STUDY ON DEEPGLOBE. IMPACT OF THE C2F-FEP, LSM-SA, AND GC-SA MODULES ON ROAD EXTRACTION PERFORMANCE. BEST RESULTS ARE HIGHLIGHTED IN BOLD

Modules			Road IoU	F1-score	Precision	Recall
C2F-FEP	LSM-SA	GC-SA	(%)	(%)	(%)	(%)
○	○	○	62.19	76.69	79.09	74.42
●	○	○	65.15	78.90	80.61	77.26
●	●	○	65.26	78.98	**81.24**	76.85
●	●	●	**65.40**	**79.08**	80.70	**77.53**

1) baseline: swin transformer without any proposed modules; *2) +C2F-FEP:* Baseline augmented with the C2F-FEP module; *3) +LSM-SA:* Configuration 2 further augmented with LSM-SA; and *4) +GC-SA:* Configuration 3 further augmented with GC-SA and the spatial-channel gate for fusion.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:30:11 UTC from IEEE Xplore. Restrictions apply.

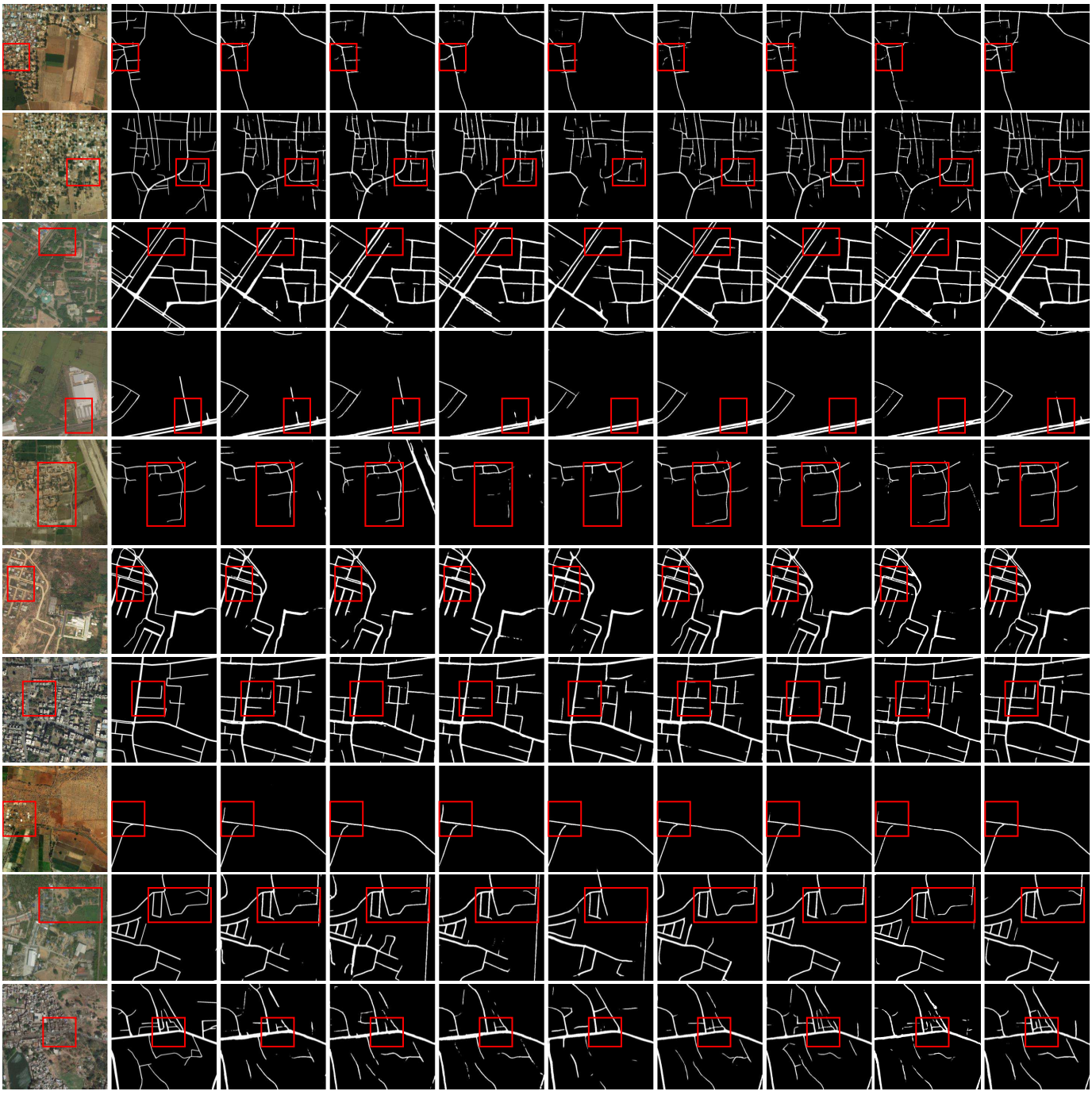


Fig. 12. Qualitative comparison on DEEPGLOBE. (a) Input image. (b) Ground truth. (c) UNet3+. (d) U-Net. (e) SegNet. (f) SETR. (g) D-LinkNet. (h) NL-LinkNet. (i) Swin transformer. (j) LGST-Net.

Fig. 11 plots the training loss curves for these four setups on both datasets, showing that each added module accelerates convergence. Quantitative results in Tables II and III confirm that our modules progressively boost road extraction accuracy.

H. Visual Comparison

Figs. 12 and 13 display qualitative results on DEEPGLOBE and ROADTRACER. Each figure presents ten samples: (a) input image, (b) ground truth, and (c)–(i) outputs from UNet3+, UNet, SegNet, SETR, D-LinkNet, NL-LinkNet, and LGST-Net. In Fig. 12 (mixed rural/urban scenes), LGST-Net yields more complete road maps with fewer artifacts and stronger connectivity at intersections. In Fig. 13 (dense urban cores), although all methods struggle, LGST-Net maintains superior structural integrity. Red boxes highlight key areas.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:30:11 UTC from IEEE Xplore. Restrictions apply.

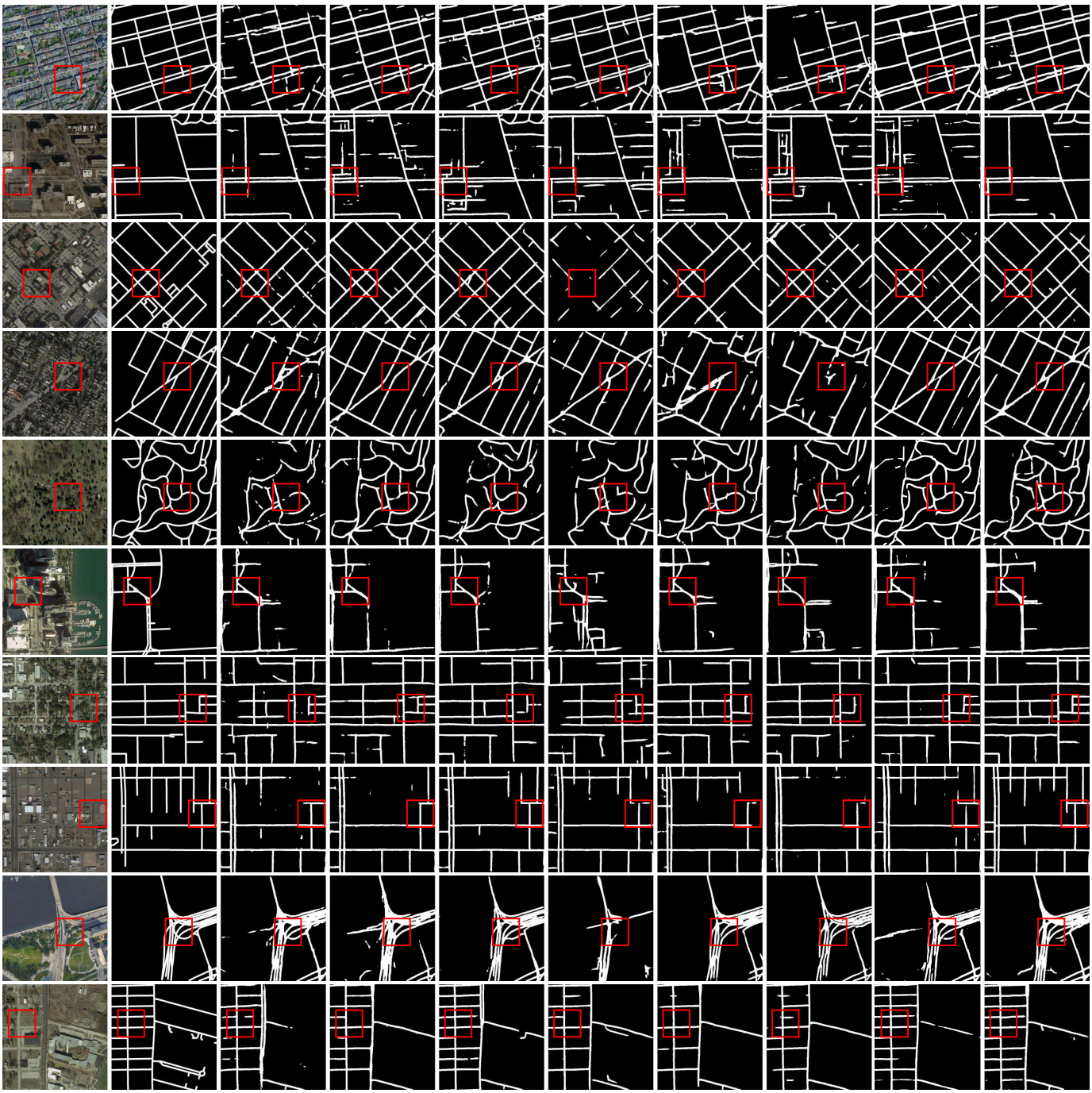


Fig. 13. Qualitative comparison on ROADTRACER. (a) Input image. (b) Ground truth. (c) UNet3+. (d) U-Net. (e) SegNet. (f) SETR. (g) D-LinkNet. (h) NL-LinkNet. (i) Swin transformer. (j) LGST-Net.

V. CONCLUSION

We introduced the LGST-Net, featuring two key modules, C2F-FEP and LGST, specifically designed for road extraction from remote sensing imagery. By combining LSM-SA and GC-SA, LGST-Net efficiently captures both fine-grained local details and long-range global context while avoiding the heavy computation of standard transformer architectures.

Extensive experiments on DEEPGLOBE and ROADTRACER datasets, compared against UNet3+, UNet, SegNet, SETR, D-LinkNet, and NL-LinkNet, demonstrate that LGST-Net achieves state-of-the-art performance with 65.40% IoU on DEEPGLOBE and 59.40% IoU on ROADTRACER. Although our sliding sparse mask exhibits strong regularity, further optimization, such as rewriting core sparse-attention kernels in Triton, can improve memory usage and computational speed. Finally, extracting roads in dense urban environments remains challenging; future work will focus on enhancing robustness and accuracy in these complex cityscapes.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:30:11 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] S. Abdullahi, B. Pradhan, and M. N. Jebur, "GIS-based sustainable city compactness assessment using integration of MCDM, Bayes theorem and RADAR technology," *Geocarto Int.*, vol. 30, no. 4, pp. 365–387, Apr. 2015.

[2] A. M. Youssef, S. A. Sefry, B. Pradhan, and E. A. Alfadail, "Analysis on causes of flash flood in jeddah city (Kingdom of Saudi Arabia) of 2009 and 2011 using multi-sensor remote sensing data and GIS," *Geomatics, Natural Hazards Risk*, vol. 7, no. 3, pp. 1018–1042, May 2016.

[3] R. Lian, W. Wang, N. Mustafa, and L. Huang, "Road extraction methods in high-resolution remote sensing images: A comprehensive review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5489–5507, 2020.

[4] A. Abdollahi, B. Pradhan, N. Shukla, S. Chakraborty, and A. Alamri, "Deep learning approaches applied to remote sensing datasets for road extraction: A state-of-the-art review," *Remote Sens.*, vol. 12, no. 9, p. 1444, May 2020.

[5] J. D. Wegner, J. A. Montoya-Zegarra, and K. Schindler, "A higher-order CRF model for road network extraction," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 1698–1705.

[6] G. Máttyus, S. Wang, S. Fidler, and R. Urtasun, "Enhancing road maps by parsing aerial images around the world," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1689–1697.

[7] Z. Zhang, Q. Liu, and Y. Wang, "Road extraction by deep residual U-Net," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 5, pp. 749–753, May 2018.

[8] W. Ding, H. Huang, and Y. Wang, "Semantic segmentation of remote sensing architectural images based on GAN and UNet3+ model," in *Proc. Pac. Rim Int. Conf. Artif. Intell.*, 2023, pp. 262–273.

[9] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[10] B. Zhou, A. Lapedriza, J. Xiao, A. Torralab, and A. Oliva, "Learning deep features for scene recognition using places database," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2014, pp. 487–495.

[11] F. Yu, Y. Zhang, H. Li, C. Du, L. Liu, and M. Jiang, "Phase contour enhancement network for clothing parsing," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 2784–2793, Feb. 2024.

[12] C. Tian, Y. Xu, and W. Zuo, "Image denoising using deep CNN with batch renormalization," *Neural Netw.*, vol. 121, pp. 461–473, Jan. 2020.

[13] X. Lu, Y. Zhong, Z. Zheng, and L. Zhang, "GAMSNet: Globally aware road detection network with multi-scale residual learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 340–352, May 2021.

[14] L. Zhou, C. Zhang, and M. Wu, "D-LinkNet: LinkNet with pretrained encoder and dilated convolution for high resolution satellite imagery road extraction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 182–186.

[15] Y. Wang, J. Seo, and T. Jeon, "NL-LinkNet: Toward lighter but more accurate road extraction with nonlocal operations," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 3000105.

[16] Z. Yang, D. Zhou, Y. Yang, J. Zhang, and Z. Chen, "Road extraction from satellite imagery by road context and full-stage feature," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, 2023, Art. no. 8000405.

[17] Y. Ren, Y. Yu, and H. Guan, "DA-CapsUNet: A dual-attention capsule U-Net for road extraction from remote sensing imagery," *Remote Sens.*, vol. 12, no. 18, p. 2866, Sep. 2020.

[18] J. Li, Y. Liu, Y. Zhang, and Y. Zhang, "Cascaded attention DenseUNet (CADUNet) for road extraction from very-high-resolution images," *ISPRS Int. J. Geo-Inf.*, vol. 10, no. 5, p. 329, May 2021.

[19] L. Gao et al., "Road extraction using a dual attention dilated-LinkNet based on satellite images and floating vehicle trajectory data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10428–10438, 2021.

[20] S. Bose, R. S. Chowdhury, D. Pal, S. Bose, B. Banerjee, and S. Chaudhuri, "Multiscale probability map guided index pooling with attention-based learning for road and building segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 206, pp. 132–148, Dec. 2023.

[21] Y. Duan, J. Qu, L. Zhang, X. Qu, and D. Yang, "LGRF-Net: A novel hybrid attention network for lightweight global road feature extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[22] Z. Wang et al., "A road-detail preserving framework for urban road extraction from VHR remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.

[23] Z. Wang, L. Xiang, Z. Liu, and Z. Wang, "Automatic extraction of roads from multisource geospatial data using fusion attention network and regularization algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.

[24] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[25] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[26] Z. Zhong, J. Li, W. Cui, and H. Jiang, "Fully convolutional networks for building and road extraction: Preliminary results," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jun. 2016, pp. 1591–1594.

[27] Y. Wei, Z. Wang, and M. Xu, "Road structure refined CNN for road extraction in aerial image," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 709–713, May 2017.

[28] R. Alshehhi and P. R. Marpu, "Hierarchical graph-based segmentation for extracting road networks from high-resolution satellite images," *ISPRS J. Photogramm. Remote Sens.*, vol. 126, pp. 245–260, Apr. 2017.

[29] T. Panboonyuen, P. Vateekul, K. Jitkajornwanich, and S. Lawawirojwong, "An enhanced deep convolutional encoder–decoder network for road segmentation on aerial imagery," in *Proc. Int. Conf. Comput. Inf. Technol.*, 2017, pp. 191–201.

[30] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[31] T. Panboonyuen, K. Jitkajornwanich, S. Lawawirojwong, P. Srestasathiern, and P. Vateekul, "Road segmentation of remotely-sensed images using deep convolutional neural networks with landscape metrics and conditional random fields," *Remote Sens.*, vol. 9, no. 7, p. 680, Jul. 2017.

[32] X. Li, H. Chen, X. Qi, Q. Dou, C.-W. Fu, and P.-A. Heng, "H-DenseUNet: Hybrid densely connected UNet for liver and tumor segmentation from CT volumes," *IEEE Trans. Med. Imag.*, vol. 37, no. 12, pp. 2663–2674, Dec. 2018.

[33] H. Huang et al., "UNet 3+: A full-scale connected UNet for medical image segmentation," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 1055–1059.

[34] Q. Zhu et al., "A global context-aware and batch-independent network for road extraction from VHR satellite imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 353–365, May 2021.

[35] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[36] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[37] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.

[38] R. Strudel, R. Garcia, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7242–7252.

[39] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 548–558.

[40] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6881–6890.

[41] T. Lu, X. Zhong, and L. Zhong, "MSwinUNet: A multi-modal U-shaped Swin transformer for supervised change detection," *J. Intell. Fuzzy Syst.*, vol. 46, no. 2, pp. 4243–4252, Feb. 2024.

[42] L. Yuan et al., "Tokens-to-token ViT: Training vision transformers from scratch on ImageNet," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 538–547.

[43] X. Chu, Z. Tian, B. Zhang, X. Wang, and C. Shen, "Conditional positional encodings for vision transformers," 2021, *arXiv:2102.10882*.

[44] K. Chen, Z. Zou, and Z. Shi, "Building extraction from remote sensing images with sparse token transformers," *Remote Sens.*, vol. 13, no. 21, p. 4441, Nov. 2021.

[45] K. Chen, C. Liu, B. Chen, W. Li, Z. Zou, and Z. Shi, "DynamicVis: An efficient and general visual foundation model for remote sensing image understanding," 2025, *arXiv:2503.16426*.

[46] S. Ren, S. Chen, S. Li, X. Sun, and L. Hou, "TESTA: Temporal-spatial token aggregation for long-form video-language understanding," in *Proc. Findings Assoc. Comput. Linguistics*, 2023, pp. 932–947.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:30:11 UTC from IEEE Xplore. Restrictions apply.

[47] W. Zeng et al., "Not all tokens are equal: Human-centric visual analysis via token clustering transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11091–11101.
[48] A. Lin, B. Chen, J. Xu, Z. Zhang, G. Lu, and D. Zhang, "DS-TransUNet: Dual Swin transformer U-Net for medical image segmentation," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–15, 2022.
[49] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.
[50] X. Liu et al., "RoadFormer: Road extraction using a Swin transformer combined with a spatial and channel separable convolution," *Remote Sens.*, vol. 15, no. 4, p. 1049, Feb. 2023.
[51] T. Chen, D. Jiang, and R. Li, "Swin transformers make strong contextual encoders for VHR image road extraction," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 3019–3022.
[52] S. Woo, J. Park, J. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 3–19.
[53] I. Demir et al., "DeepGlobe 2018: A challenge to parse the Earth through satellite images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 172–181.
[54] F. Bastani et al., "RoadTracer: Automatic extraction of road networks from aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4720–4728.
[55] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput. Assist. Interv.*, 2015, pp. 234–241.
[56] G. Zhou, W. Chen, Q. Gui, X. Li, and L. Wang, "Split depth-wise separable graph-convolution network for road extraction in complex environments from high-resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.
[57] R. Zuo, G. Zhang, R. Zhang, and X. Jia, "A deformable attention network for high-resolution remote sensing images semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.
[58] L. Dai, G. Zhang, and R. Zhang, "RADANet: Road augmented deformable attention network for road extraction from complex high-resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.



Xu Wang received the B.S. degree in software engineering from the School of Software Engineering, Huazhong University of Science and Technology, Wuhan, China, in 2014, and the Ph.D. degree in communication and information systems from the School of Computer Science, Wuhan University, Wuhan, in 2020.

He is currently an Associate Professor with the School of Information and Communication, National University of Defense Technology, Wuhan. His research interests include video coding, image processing, cross-modal retrieval, and remote sensing.



Luyao Wang received the B.S. degree in geomatics engineering from the School of Geodesy and Geomatics, Wuhan University, Wuhan, China, in 2015, and the Ph.D. degree in geography from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, in 2021.

He was a Visiting Student with the Department of Urban Studies and Planning, Massachusetts Institute of Technology, Cambridge, MA, USA, from 2019 to 2020. He is currently an Associate Professor with the School of Information and Communication, National University of Defense Technology, Wuhan. His research interests include urban geography, nighttime light data applications, sustainable development, and consumer activities.



Wenxin Huang (Member, IEEE) received the B.S. degree in information technology from Metropolia University of Applied Sciences, Helsinki, Finland, in 2010, the M.S. degree in geoinformatics from Aalto University, Helsinki, in 2012, and the Ph.D. degree in computer science from Wuhan University, Wuhan, China, in 2020.

She is currently a Lecturer with Hubei University, Wuhan, and a Research Officer with the Centre for Frontier AI Research, Agency for Science, Technology and Research (A*STAR), Singapore. Her research interests include multimedia content analysis and retrieval, large-scale multimedia data mining, and artificial intelligence.



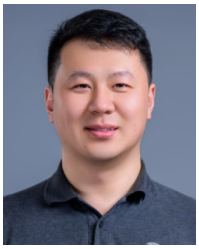
Yibo Liu received the B.S. degree in computer science from Xi'an University of Posts and Telecommunications, Xi'an, China, in 2012, and the Ph.D. degree from the National University of Defense Technology, Changsha, China, in 2024.

He is currently a Lecturer with the College of Information and Communication, National University of Defense Technology, Wuhan, China. His research interests include deep learning, computer vision, and knowledge engineering.



Sheng Zhang received the B.S. degree in computer applications from PLA University of Science and Technology, Nanjing, China, in 2001, and the Ph.D. degree in military communication from the National Defense University of Information Technology, Wuhan, China, in 2012.

He is currently an Associate Professor with the School of Information and Communication, National University of Defense Technology. His research interests include cross-modality analytics, image processing, and data engineering.



Xian Zhong (Senior Member, IEEE) received the B.S. degree in Computer Science from Wuhan University, Wuhan, China, in 2007, and the Ph.D. degree in Computer Science from the Huazhong University of Science and Technology, Wuhan, in 2013.

He was a Visiting Scholar with the Peking University, Beijing, China, from 2021 to 2022, and ROSE@EEE, Nanyang Technological University, Singapore, in 2024. He is currently a Professor with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan. His research interests include multimodal information processing, neuromorphic vision, and intelligent transportation systems.

Dr. Zhong serves as an Area Chair for IJCAI, ICME, ICIP, and IJCNN. He serves as an Associate Editor for TVC and VRIH.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:30:11 UTC from IEEE Xplore. Restrictions apply.